

Signal Generators from some manufacturers

Sciencetech Technologies Pvt. Ltd.	GW Instek	Rohde & Schwarz India Pvt Ltd	Tektronix	Agilent
Model Sciencetech 4211 1mHz-10MHz Function/Arbitrary Waveform Generator	AFG-2225 True Dual-Channel Arbitrary Function Generator	Model	AFG2021	Agilent 33512B Waveform Generator
Sine, square, triangle, ramp, pulse and more than 25 arbitrary waveforms	Sine, square, ramp, pulse, noise, ARB	Sine, square, pulse, triangle, ramp, arbitrary waveforms incl. standard curves (white noise, cardiac, etc)	Sine, square, pulse, ramp, noise, DC, Sin(x)/x, Gaussian, Lorentz, exponential rise, exponential decay and Haversine	Standard: Sine, square, ramp, pulse, triangle, Gaussian noise, PRBS (Pseudorandom Binary Sequence), DC Built-in arbitrary: Cardiac, exponential fall, exponential rise, Gaussian pulse, Haversine, Lorentz, D-Lorentz, negative ramp, sinc
1 mHz to 10 MHz	1 μ Hz to 25 MHz	10 μ Hz to 25 MHz	1 μ Hz -20 MHz	20MHz
20Vpp (0.C), 10Vpp (50 Ω)	1 mVpp - 10Vpp (into 50 Ω) 2 mVpp - 20Vpp (open-circuit)	5 mVpp - 10 Vpp	10 mVpp - 10Vpp (50 Ω), 20 mVpp - 20Vpp (open circuit)	1 mVpp - 10 Vpp (50 Ω)
<0.3%	$\leq$ -55dBc	0.04% (f <100 kHz)	<0.2% (10 Hz to 20 kHz, 1Vp-p)	<0.04%
Digitally synthesised amplitude, no external attenuation required	NA			NA
50 Ω	50 Ω	50 Ω	50 Ω	50 Ω
<20 ns	$\leq$ 25 ns	<8 ns (square) 8 ns to 500 ns variable-edge-time (pulse)	$\leq$ 20 ns	100 ps
100 MSA/s	120 MSA/s	250 MSA/s, 14-bit, 256 kpts	250 MSA/s	1 μ SA/s to 250 MSA/s
4096	4k points			up to 16 MSA/channel
$\pm$ 9.5V at amplitude 1Vpp (amplitude + DC offset = 20Vpp max.)	$\pm$ 5 Vpk AC + DC (into 50 ohms); $\pm$ 10Vpk AC + DC (open-circuit)	$\pm$ 5 mV to 5V (into 50 Ω) $\pm$ 10 mV to 10V (open-circuit)	$\pm$ (5Vpeak – amplitude Vp-p/2) (50 Ω load), $\pm$ (10Vpeak – amplitude Vp-p/2) (open-circuit)	$\pm$ (5 VDC - Peak AC) into 50 Ω $\pm$ (10 VDC - Peak AC) into open circuit
Yes	2 mVpp - 20Vpp (open-circuit)	10 mVpp-20Vpp (open-circuit)		Amplitude range: 2 mVpp-20 Vpp into open circuit DC offset: $\pm$ (10 VDC - Peak AC) into open circuit
Linear sweep with user-defined start frequency, stop frequency and time of sweep	1 ms to 500 s	Signals: All (except pulse) Type: Linear/logarithmic Direction: up/down Sweep time: 1 ms to 500 sec		Linear, Logarithmic, List (up to 128 user-defined frequencies)
RS232	USB (host & device)	USB/RS-232 dual-interface, optional Ethernet/USB Dual-interface or IEEE-488 (GPIB)		LAN(LXI-C), USB(Host and Device) and GPIB
NA	Time base accuracy $\pm$ 1 count			NA
320x240 TFT true-colour LCD	8.89cm (3.5 inch) TFT LCD	8.89cm (3.5 inch) colour TFT QVGA 65k colours		10.9cm (4.3 inch) Color TFT WQVGA (480x272) with LED backlight
26.2x13x31.6 cm ³	26.6x10.7x29.3 cm ³	28.5x7.5x36.5 cm ³		2U x 1/2 rack width
Based on Direct Digital Synthesis (DDS) technique, pulse duty cycle from 5% to 95%, internal modulation - AM (variable depth of modulation), FM (variable deviation frequency), ASK/burst (variable number of cycle and time), FSK (variable F1, F2 and hop frequency), waveform creator software for custom waveform generation, 320x240 TFT true-colour LCD for simultaneous parameter display, non-volatile storage & recall for control panel setting	True dual-channel output, CH2 provides the same characteristics as CH1; built-in AM/RM/PM/FSK/SUM/sweep/burst and frequency counter; couple, tracking, phase operations of dual channel are supported; 1% – 99% adjustable duty cycle for square waveform	Modulations like AM, FM, PM, PWM, FSK can also be generated		Superior signal integrity with Trueform technology providing highest resolution and lowest distortion Superior user experience with intuitive front panel, complimentary software and localized help and manuals, and access to manuals from mobile devices • Get advanced signal creation/editing capability without tedious programming with 33503A BenchLink Waveform Builder Pro and Basic Software • Create a waveform from the front panel using the built-in editor • Capture a waveform from an oscilloscope and download it to the generator • Create a waveform in MATLAB, Microsoft® Excel, etc. and download it to the generator